– RESULTS · AUTO-FRAMING –

One fixed camera in, a performer-focused shot out

BEFORE · raw fixed camera



AFTER · StageLens



Left: raw footage. Right: StageLens output. Same clip, no operator.

1.1×–2.3×
virtual zoom, capped at 2.5×

5–6 dancers
kept fully in frame

0 cut heads
in 40 sampled frames

REAL STAGE FOOTAGE · 1080P 60 FPS · CPU PROTOTYPE

– RESULTS · ROBUSTNESS –

Works across lighting setups, and we know where it stops

Clip	Content	Dancers	Zoom	FPS (CPU)
A	warm / white spotlights	5	2.14×	5.8
C	blue light	6	2.15×	4.6
D	purple backlight, dark outfits	7	1.97×	5.3
B	8 → solo + 2 kneeling	8 → 3	1.48×	5.7

Zoom = median virtual zoom per clip. B is the current limit: side performers keep the crop wide.

Known limits

- Crowds wider than 80% of the stage fall back to a full frame
- A lead performer among side performers is not picked yet → Stage II
- Lighting restoration needs over-exposed faces; effect is small on wide stage shots

TESTED ON 4 CLIPS · WARM SPOTS, BLUE LIGHT, PURPLE BACKLIGHT

– RESULTS · PERFORMANCE –

Detection is ~95% of the time: the part UGen300 takes over

Throughput

5.7 FPS

CPU prototype, 1080p

target ≥ 30 FPS on UGen300

Latency (frames with detection)

~300 ms

median, CPU

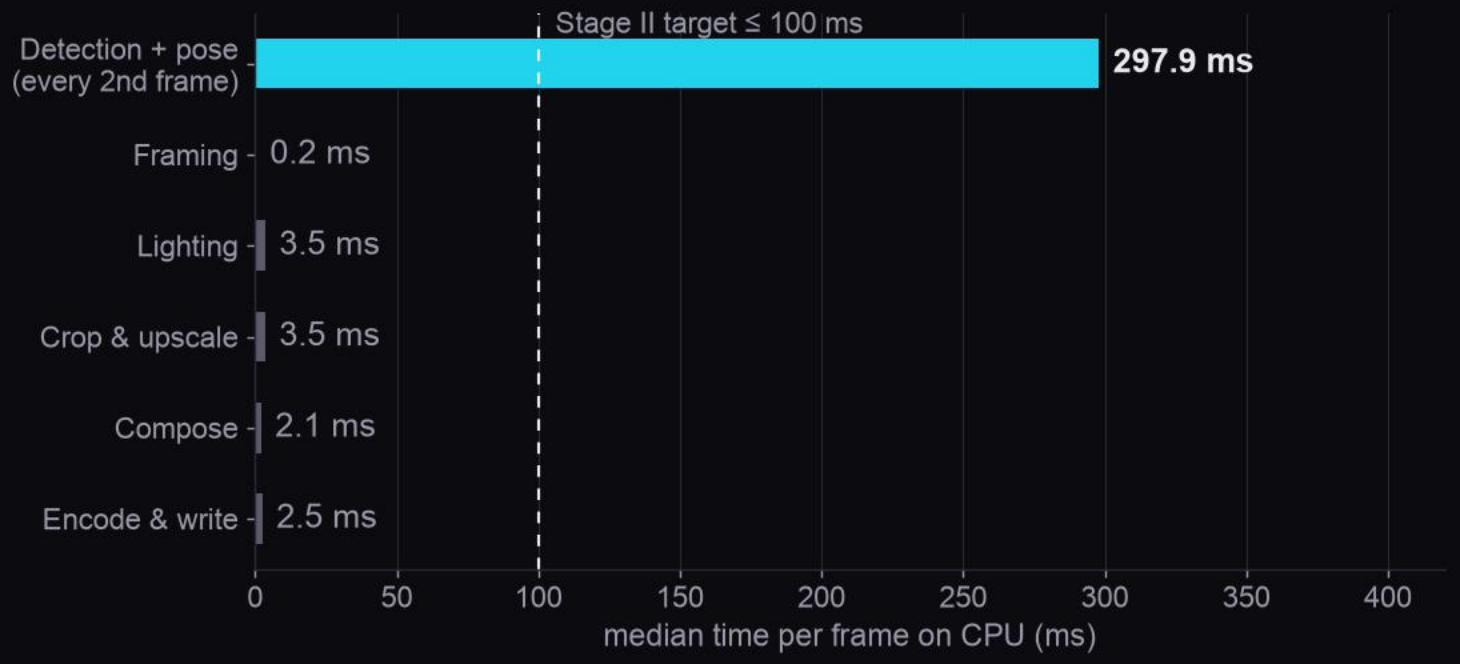
target ≤ 100 ms end-to-end

Auto-framing

1.1×–2.3×

zoom range, 4 test clips

40 sampled frames: 0 cut heads



Stage II target

≥ 30 FPS

≤ 100 ms

measured on UGen300

Pose inference every 2nd frame is the bottleneck. Moving it to the NPU frees the CPU for framing and lighting.

All numbers measured on a CPU prototype; UGen300 results will be added in Stage II.